GolfBot: Autonomous robot for Divot Detection and Repair

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Abstract

Maintaining the pristine condition of golf course turf is a significant operational challenge, demanding constant, labor-intensive efforts to repair surface damage such as divots. This manual process is not only costly and time-consuming but can also lead to inconsistent repairs, impacting the quality of play. The advancement of autonomous systems presents a compelling opportunity to automate such specialized maintenance tasks, promising increased efficiency, precision, and a reduction in operational costs.

This project presents the design, implementation, and evaluation of GolfBot, a functional prototype of an autonomous rover created to address the specific task of divot repair. The system is built upon the Wumpus mobile robot platform and enhanced with a suite of specialized hardware and software. Its core functionalities are driven by a computer vision system employing a custom-trained YOLO model for real-time divot detection, an RTK-GPS module for centimeter-level navigation accuracy, and a custom-designed mechanical dispenser for the precise application of a sand-seed mixture. The entire system is integrated and controlled using the Robot Operating System 2 (ROS 2), with key components developed in Python and C++, while a custom GUI provides tele-operation and mission monitoring.

The result of this project is a functional prototype capable of autonomously navigating a designated area, identifying divots, and performing targeted repairs. This thesis documents the complete development lifecycle, from the initial hardware and software architecture design to the implementation of the core operational logic and field testing. The evaluation of the prototype demonstrates the viability of using autonomous robotics to improve the efficiency and quality of golf course maintenance, establishing a solid foundation for future enhancements and potential commercial application. GolfBot detected divots with an mAP_{50} of 0.91, localised repairs within 2.3 ± 0.8 cm (1σ) of ground-truth RTK survey points, and dispensed material with a

volumetric error below 8 %.

The result of this project is a functional prototype of an autonomous divot-repairing robot. This thesis documents the design, implementation, and evaluation of the GolfBot, demonstrating its capability to improve the quality and efficiency of golf course maintenance.

Declaration

I hereby declare that this thesis is entirely my own work and has not been submitted for any other degree or professional qualification. All sources I have used or quoted have been indicated and acknowledged by means of complete references.

The work described herein was conducted as part of the Master of Science program at Maynooth University. The GolfBot project was developed using university resources and facilities. As such, any intellectual property arising from this project is subject to the policies of Maynooth University.

Acknowledgments

I would like to extend my sincere gratitude to my thesis supervisor, Professor Gerard Lacey. His insightful advice, consistent guidance, and unwavering support were invaluable in navigating my progress throughout this project. His expertise helped shape the direction of this research, and I am deeply appreciative of his mentorship.

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I am also grateful to Cecil Ryan, who provided me with the invaluable opportunity to work as a software developer intern at his company. This professional experience enhanced my software development skills and offered crucial insights into teamwork and project management, which greatly benefited my approach to this thesis.

Finally, I wish to thank my family and friends for their encouragement and support throughout this journey.

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Acronyms

DoF Degree(s) of Freedom. xi

GUI Graphical User Interface. xi, 4

ROS Robot Operating System. xi, 1, 3

RTK Real-Time Kinematic. xi, 1

YOLO You Only Look Once. xi

YOLOv11-Seg You Only Look Once v11-Seg. xi

Chapter 1

Introduction

Maintaining the immaculate turf of a golf course is a relentless and costly endeavor. Among the most repetitive tasks is the repair of divots—patches of turf displaced by golf swings—which, if left unattended, disrupt the playing surface and detract from the course's quality. This thesis introduces the GolfBot project, an autonomous robotic solution designed to automate the detection, localization, and repair of these divots. This chapter introduces the motivation and problem statement that drive the project. It then details the core components of the GolfBot system and provides an outline of the subsequent chapters, serving as a roadmap for the reader.

1.1 Motivation

The primary motivation for this project arises from a distinct gap in the market for automated golf course maintenance. Divots are inevitable on any busy golf course. An average 18-hole course needs a tons of sand-seed mixture to fill the divots during peak season. Current practice relies on green-keeping staff walking the fairways and manually filling cavities with a sand–seed mixture — a task that is repetitive, physically demanding, and subject to human inconsistency. Meanwhile, the robotics industry has delivered commercial mowing and fertilizer-spreading robots, but no solution that identifies and repairs individual divots. Advances in edge AI (e.g. Nvidia Jetson-class GPUs), low-cost Real-Time Kinematic (RTK), and open-source mobile-robot frameworks (Robot Operating System (ROS)) running on a Linux based system have made it feasible to develop a specialized, cost-effective prototype to automate this precise task.

1.2 Problem Statement

The project addresses the following core problem: *Create a mobile system that can: (1) automatically identify turf divots in real time on an outlined section of a golf course fairway; (2) navigate to each detected divot with centimeter accuracy; and (3) dispense an appropriate amount of sand-seed mix to fill the divot and restore the playing surface — all without the need for human intervention.* Key technical challenges include:

- Visual detection of irregular, depressions under variable outdoor lighting.
- Robust localisation across tens of hectares with minimal GNSS occlusion.
- Precise metering of a sand–seed mixture.
- Real-time coordination of perception, planning, and actuation on a battery-powered embedded computer.

1.3 Objectives and Project Scope

The main objective of this project is to design, implement, and evaluate a working prototype a proof-of-concept, autonomous robot, capable of divot repair.

The scope of the project encompasses the following key activities:

- A review of the current state-of-the-art in autonomous turf maintenance, computer vision techniques for object detection in natural environments, and automated dispensing mechanisms.
- Collect and annotate a minimum 1 500-image dataset of divots for model training.
- Achieve real-time inference ($\leq$40 ms per frame) on embedded GPU hardware.
- The design and integration of the complete hardware and software architecture required for the robot's operation.
- The implementation and testing of the system's principal components: the computer vision pipeline, the high-precision navigation system, and the mechanical repair dispenser.

- A thorough validation of the prototype's performance in both controlled and real-world scenarios to assess its effectiveness.
- A critical discussion of the prototype's current limitations and the identification of potential avenues for future research and development.

1.4 The GolfBot System

The GolfBot is a custom-built autonomous robot designed to meet the specific demands of divot repair. The system is built upon the Wumpus robot platform, which was developed at Maynooth University and served as the project's foundation. This base platform was significantly modified and enhanced with a suite of specialized hardware to achieve the project's goals.

The final system integrates three primary subsystems working in parallel:

- A **computer vision system** for detecting and localizing divots in real-time.
- A **high-precision navigation system**, using RTK-GPS, to guide the robot to target locations with centimeter-level accuracy.
- A **mechanical dispensing system** to deliver a controlled amount of repair material into the divot.

These capabilities are powered by an NVIDIA Jetson Orin Nano, an Intel RealSense depth camera, and a custom-designed dispenser, all managed within the ROS framework. The detailed account of the platform's initial condition and the full scope of modifications and hardware integrations are presented in Chapter 4.

1.5 Thesis Outline

The following chapters will detail the project.

- **Chapter 2: Literature Review** will survey existing work on autonomous robots in agriculture and turf maintenance, divot detection techniques, and automated repair systems.
- **Chapter 3: Design** will present the overall design of the GolfBot, covering both the hardware and software architectures.

- **Chapter 4: Implementation** will provide a detailed description of the implementation of the GolfBot's main components: the computer vision system, the navigation system, the mechanical dispenser, and the Graphical User Interface (GUI).
- **Chapter 5: Experimental Results** will describe the experiments conducted to validate the performance of the GolfBot and present the results.
- **Chapter 6: Future Work** will discuss potential enhancements and future research directions for the GolfBot project.
- **Chapter 7: Conclusion** will summarize the project, highlighting its contributions and concluding remarks.

Chapter 2

Literature Review

2.1 Autonomous Robots in Turf Maintenance

2.2 Divot Detection and Analysis

2.3 Automated Repair Mechanisms

2.4 Summary

Chapter 3

Design

3.1 Overall Design

The design of the GolfBot is centered around creating a robust and functional platform capable of navigating semi-structured outdoor environments while performing its specialized repair task. The system's design can be broken down into three core areas: the physical mechanical structure, the electronic hardware architecture, and the software architecture that governs its behavior. This chapter details the design decisions made in each of these areas. Figure 3.1 provides a visual overview of the completed robot.

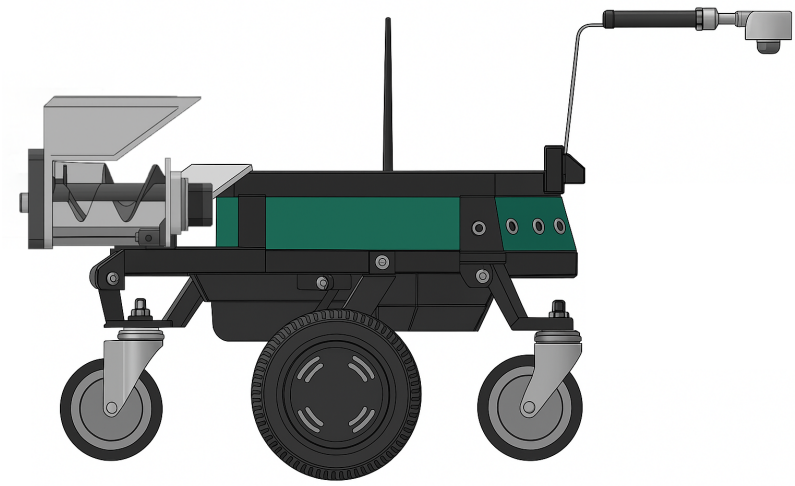

Figure 3.1: GolfBot System Diagram

3.2 Mechanical Design

The mechanical design focuses on the robot's physical construction, ensuring stability, durability, and the proper placement of all sensors and actuators.

3.2.1 Chassis and Drivetrain

The robot's structure is based on a lightweight aluminum chassis with main body dimensions of 500 mm x 360 mm x 115 mm. It uses a differential drive architecture, powered by two wheel hub motors with a wheel diameter of 165 mm and a track width (distance between the motors) of 570 mm. For stability, the drivetrain is supported by rubber caster wheels.

The motor hubs and casters are mounted on two swing arms, which function as a simple suspension system. This allows the robot's main body to remain relatively horizontal while the swing arms absorb minor bumps from uneven terrain. The wheels provide a ground clearance of 100 mm, sufficient for overcoming common obstacles on a golf course fairway. To protect the electronics, all internal components are housed within the main body frame, secured in custom 3D-printed holders and attached with strong, double-sided Velcro to prevent movement during operation.

3.2.2 Sensor and Component Mounting

To ensure optimal data collection, custom mounts were designed for the primary sensors. An L-shaped 25 mm aluminum box section is mounted on top of the robot's aluminum plate. This structure holds the camera 300 mm in front of the robot and 600 mm above the ground, providing a clear and effective field of view. To protect the camera from direct sunlight and physical impacts, it is housed in a 3D-printed holder featuring a 90-degree hinge, which ensures the camera is positioned orthogonally to the ground.

The same aluminum arm also supports a 2 mm thick stainless steel circular plate that serves as a ground plane for the RTK-GPS rover antenna. Additionally, a stand for the LCD screen, used for the graphical user interface, is mounted on the top frame.

3.2.3 Dispensing and Repair Mechanism

The repair system is mounted at the rear of the robot. A 90-degree folded stainless steel bracket, which was laser-cut for precision, holds the sand dispenser 100 mm above the ground and approximately 900 mm behind the camera's center point. The dispenser itself is 3D-printed from PLA plastic. Its core is a screw auger mechanism, powered by a NEMA 17 stepper motor, which pushes the sand-seed mixture from a hopper out through a nozzle.

Mounted directly behind the dispenser nozzle is a rubber-mounted brush. This brush extends 60 mm from the nozzle and serves to evenly spread the dispensed material, ensuring the divot is filled smoothly. It is attached to the same stainless steel bracket using custom 3D-printed mounts.

3.2.4 RTK Base Station Design

The ArduSimple RTK system necessitates a stationary base station to deliver correction data to the rover. To achieve this, the base module is enclosed within a waterproof plastic box and mounted on a metal pipe. A circular, laser-cut metal plate is affixed to the top of the pipe to support the RTK base antenna. This entire setup is meticulously designed to remain stationary, ensuring signal integrity by positioning it at the highest feasible point within the robot's operational area. The base station is powered through a standard USB cable, which can be connected to a laptop or a simple USB wall adapter. As shown in Figure 3.2, the base antenna setup is depicted here.

Figure 3.2: RTK Base Antenna Setup

3.3 Hardware Architecture

The hardware architecture describes the electronic components and the flow of data between them. It is designed around a central, powerful controller that processes sensor data and sends commands to the various actuators.

The Arduino Uno plays a critical role as a dedicated low-level controller. It is responsible for managing the NEMA 17 stepper motor (via the A4988 driver) for the dispensing mechanism and reading orientation data from the BNO055 IMU sensor. The Arduino communicates with the main Jetson Orin controller over a serial-over-USB connection. The Rover RTK-GPS module is also connected directly to the main controller via USB, providing it with high-precision location data. This separation of concerns allows the main controller to focus on high-level processing while the Arduino handles real-time actuator control and sensor reading.

3.4 Software Architecture

The software architecture is designed using the ROS 2, a framework chosen for its modularity and robust communication system. The architecture's design follows a standard robotics pattern of perception, processing, and actuation, which means that different functions are handled by separate processes (nodes) that talk to each other through clear channels (topics).

This is illustrated in the conceptual diagram in Figure 3.3.

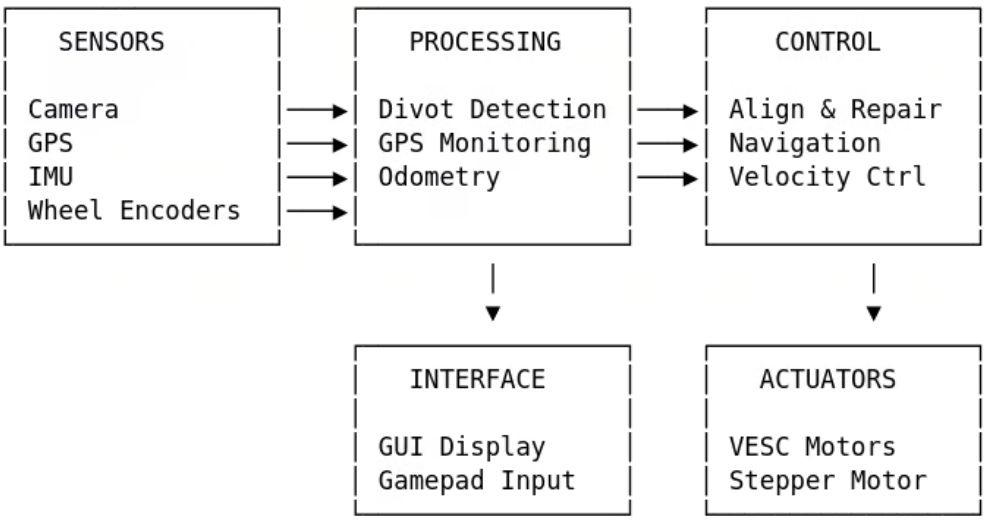

Figure 3.3: Conceptual data flow of the GolfBot system.

Figure 3.4 shows the ROS 2 Graph (rqt_graph) output from the running GolfBot system. The ellipses represent ROS 2 nodes (the processes), while the rectangular boxes represent topics (the data streams). The arrows indicate the direction of data flow, showing which nodes are publishing and which are subscribing.

The following subsections describe the roles of these key components as shown in the graph.

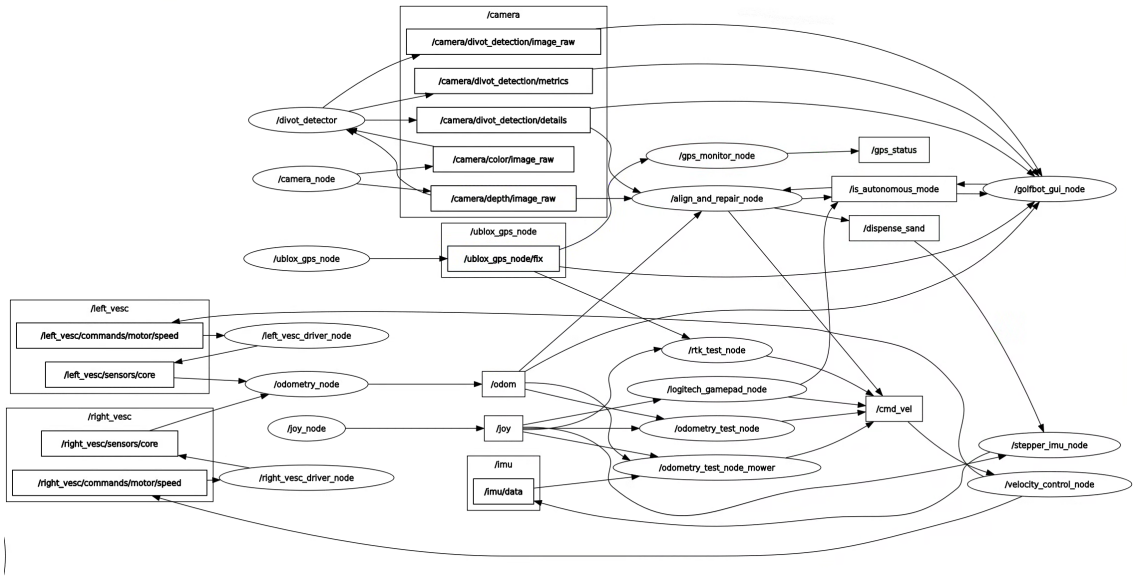

Figure 3.4: The ROS 2 computation graph (`rqt_graph`) of the running system.

3.4.1 Sensor Interface and Drivers

These nodes provide the direct interface to the robot's hardware sensors, converting physical signals into data streams for the rest of the system.

Camera Node (`/camera_node`) The driver for the Intel RealSense camera. It publishes the raw color and depth image streams to the `/camera/color/image_raw` and `/camera/depth/image_raw` topics.

GPS Driver Node (`/ublox_gps_node`) Interfaces with the U-blox RTK-GPS receiver, publishing standard `NavSatFix` messages containing latitude, longitude, and accuracy information to the `/ublox_gps_node/fix` topic.

IMU Interface Node (`/stepper_imu_node`) This node has a dual role. In its sensing capacity, it reads orientation data from the BNO055 sensor (via an Arduino) and publishes it to the `/imu/data` topic. The other role is to act as an actuator, receiving commands from the `/dispense_sand` topic and sending

them to the Arduino to activate the stepper motor and dispense the sand-seed mixture.

Joystick Driver (`/joy_node`) A standard ROS 2 node that reads the raw axis and button states from the connected gamepad and publishes them to the `/joy` topic.

3.4.2 Perception and Localization

This subsystem is responsible for processing raw sensor data to build an understanding of the environment and the robot's position within it.

Divot Detector Node (`/divot_detector`) The primary perception processing node. It subscribes to `/camera/color/image_raw` and uses an AI model to identify divots. It publishes its findings to several key topics: `/camera/divot_detection/image_` for a visualized feed, and `/camera/divot_detection/details` containing structured data on divot locations and sizes.

Odometry Node (`/odometry_node`) This node calculates the robot's movement based on wheel speeds. It subscribes to sensor data from the VESC driver nodes and publishes an estimate of the robot's pose and velocity to the `/odom` topic.

3.4.3 Control and Navigation

These nodes constitute the robot's "brain," making decisions for both manual and autonomous operation.

Manual Control Node (`/logitech_gamepad_node`) Translates the abstract messages from the `/joy` topic into concrete `geometry_msgs/Twist` messages, which contain linear and angular velocity commands. These are published on the `/cmd_vel` topic for manual driving.

Autonomous Logic Node (`/align_and_repair_node`) This node contains the logic for the autonomous "align and repair" test. It uses data from the odometry and divot detection topics to navigate the robot and sends a command to the `/dispense_sand` topic once it is correctly positioned over a divot.

Velocity Control Node (`/velocity_control_node`) Acts as a multiplexer for motion control. It takes velocity commands from the `/cmd_vel` topic (which can be published to by either the manual or autonomous node) and translates them into low-level speed commands for the VESC drivers.

3.4.4 Low-Level Actuation and User Interface

This final group includes the nodes that execute physical actions and provide a window into the system for the operator.

Motor Driver Nodes (`/left_vesc_driver_node`, `/right_vesc_driver_node`) These nodes are the final link to the drive motors. They receive speed commands from the `/velocity_control_node` and send the appropriate signals to the VESC motor controllers. They also publish sensor feedback used by the `/odometry_node`.

Dispenser Actuator Node (`/stepper_imu_node`) In its actuator capacity, this node subscribes to the `/dispense_sand` topic. When it receives a command, it sends a signal to the Arduino to activate the stepper motor and dispense the repair mixture.

GUI Node (`/golfbot_gui_node`) Provides the main graphical user interface. It subscribes to numerous topics (e.g., camera feeds, odometry, GPS status) to give the operator a complete overview of the robot's state. It can also publish commands, such as setting the `/is_autonomous_mode` flag to start an autonomous run.

Chapter 4

Implementation

4.1 Wumpus Platform Robot

The GolfBot's development started with the Wumpus platform, a pre-existing robot that provided a solid mechanical and electrical base. This section details the initial specifications of the platform, its condition as received, and the critical modifications and improvements that were made.

4.1.1 Initial Design and Specifications

The Wumpus robot, developed at Maynooth University, is a differential-drive robot. In its original configuration, it was controlled by a Raspberry Pi 4 running ROS 2. The power system was built around a 36V battery, which supplied two VESC motor controllers for the brushless DC motors. The platform also included three DC-DC converters: two specifically for the VESC controllers, and a third to step the voltage down to 12V and 5V for the main controller and other peripheral electronics.

4.1.2 Base Platform Condition

Upon initial inspection, the platform had two key issues that needed to be addressed. First, one of the VESC controllers had a recurring power-up problem. To ensure reliable operation, I installed a dedicated kill switch to control its power flow directly. Second, the motor settings were configured for high-speed performance (minimum 900 ERPM), which made the robot uncontrollable at the low speeds required for divot inspection and repair. I used the VESC tool, an open-source configuration

software, to lower the minimum motor speed to 200 ERPM. This adjustment was crucial for enabling the precise, low-speed maneuvers needed for the project.

4.1.3 Modifications and Improvements

After resolving the baseline issues, I performed significant upgrades to equip the robot for its mission. The underpowered Raspberry Pi 4 was replaced with a Seeed reComputer J4012, which features a powerful NVIDIA Jetson Orin 16GB module suitable for edge AI computing. I then integrated the following key hardware components:

- An Intel RealSense D435i stereo camera for visual and depth data.
- An Ardusimple SimpleRTK2B RTK GPS module for high-accuracy localization.
- An Arduino Uno microcontroller to manage the dispenser motor and IMU.
- A BNO055 IMU sensor for orientation tracking.
- An A4988 stepper motor driver and a NEMA 17 stepper motor to actuate the dispenser.
- A custom-designed and 3D-printed auger dispenser for the sand-seed mixture.

These enhancements fundamentally transformed the general-purpose Wumpus platform into the specialized GolfBot.

4.2 Computer Vision System

4.2.1 Introduction/Purpose

4.2.2 Data Set Collection and Preprocessing

4.2.3 Model Training

4.2.4 Hardware

4.2.5 Software

4.2.6 Implementation in ROS

4.3 Navigation System

4.3.1 Introduction/Purpose

4.3.2 Hardware Used

4.3.3 Software Implementation

4.4 Mechanical Dispenser

4.4.1 Introduction/Purpose

4.4.2 Hardware Design and 3D-Printed Components

Here I can specify the output rate of the dispenser and the volume of the sand-seed mixture that is dispensed.

4.4.3 Software Integration in ROS

4.5 Graphical User Interface (GUI)

4.5.1 Introduction/Purpose

4.5.2 Software and Framework

4.5.3 Functions, Features, and Control

Chapter 5

Experimental Results

5.1 Computer Vision System Evaluation

5.1.1 Divot Detection Accuracy

5.1.2 Real-World Performance

5.2 Navigation System Evaluation

5.2.1 Positional Accuracy

5.2.2 Path Following and Search Algorithm Performance

5.2.3 Limitations

5.3 Mechanical Dispenser Evaluation

5.3.1 Dispensing Accuracy and Consistency

5.4 GUI Evaluation

5.4.1 Usability and User Feedback

5.5 Overall Performance of the GolfBot Prototype

Chapter 6

Future Work

6.1 Model Enhancements

6.1.1 Expanding the Dataset

6.2 Hardware Enhancements

6.2.1 Enhanced Visual System

6.2.2 Improved Drivetrain

6.2.3 Upgraded Dispenser

6.2.4 Autonomous Recharging

6.2.5 Enhanced User Interface

6.2.6 Automated Refilling Stations

Chapter 7

Conclusion

7.1 Summary of Work

7.2 Contributions

Appendix A

Name

Bibliography